Helix Industries

Reactor Operations & Safety Logic – Operator Guide

> **Document Type:** Internal Operator Reference
>
> **Audience:** Control Room Operators / Maintenance Operators
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> **System:** Helix Reactor Control System (OPC UA–based PLC)

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1. Overview

This document describes the **normal operational behavior**, **safety mechanisms**, and **maintenance control logic** of the Helix Reactor system. It is intended to help operators understand:

- How reactor temperature and pressure evolve during operation
- How safety trips are triggered and latched
- Under which conditions a trip can be reset
- How the maintenance operating window is reached safely

The system is designed so that **safety always has priority** over operator control.

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2. Core Reactor Variables

The following variables are continuously monitored and enforced by the PLC:

Reactor Process Variables

- **Temperature** – Current reactor temperature (°C)
- **Pressure** – Current reactor pressure (bar)
- **CalibrationOffset** – Maintenance-only adjustment applied to sensor calibration

  CalibrationOffset does **not** directly control temperature or pressure. Instead, it introduces a controlled bias used during diagnostics and maintenance.

Safety Variables

- **TripActive** – Indicates that the reactor has entered a safety shutdown state
- **RodsInserted** – Status of safety rods (controlled automatically)
- **EmergencyCooling** – Emergency cooling system status (automatic)

Control Variables

- **Mode** – Reactor operating mode (`NORMAL` / `MAINTENANCE`)
- **TestOverride** – Enables limited maintenance overrides
- **ResetTrip** – Operator request to clear a safety trip

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3. Normal Operating Mode

In **NORMAL** mode:

- CalibrationOffset is expected to be **0.0**
- TestOverride must be **disabled**
- Reactor behavior follows standard control curves

Any attempt to apply calibration offsets or overrides in NORMAL mode is ignored by the PLC.

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4. Safety Trip Logic

A **safety trip** is automatically triggered when reactor conditions exceed hard safety limits:

Trip Thresholds (Internal)

- Temperature ≥ **\~305°C**
- Pressure ≥ **\~75 bar**

When a trip occurs:

- `TripActive` becomes **TRUE**
- Control logic is locked
- Safety systems take precedence
- Operator inputs are restricted

  Once triggered, a trip is **latched** and cannot be cleared immediately.

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5. Trip Reset Conditions

A safety trip **cannot** be reset arbitrarily.

The PLC will only accept a reset if **all** of the following conditions are met:

- Reactor temperature is below **\~288°C**
- Reactor pressure is below **\~70 bar**
- Operating mode is `NORMAL`
- TestOverride is **disabled**
- CalibrationOffset is reset to **0.0**

Only when the system is back in a verified safe state will a `ResetTrip` request be honored.

This ensures that operators cannot bypass safety systems while unsafe conditions persist.

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6. Maintenance Mode & Safety Window

Entering Maintenance Mode

Maintenance operations require explicit operator action:

1. Switch `Mode` to **MAINTENANCE**
2. Enable `TestOverride`
3. Begin controlled adjustment using `CalibrationOffset`

In this mode, the reactor is still protected by safety logic, but limited overrides are permitted for diagnostics.

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7. Maintenance Operating Window

The PLC defines a **maintenance operating window** where certain diagnostic tools become available.

This window opens when:

- Temperature reaches approximately **295°C OR Pressure 73 bar
- Pressure & Temp remains below trip thresholds
- No safety trip is active

This window exists **below** trip limits but **above** normal operating conditions.

> The window is intentionally narrow to ensure maintenance actions are time-limited and closely monitored.

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8. Behavior During CalibrationOffset Ramp

When CalibrationOffset is increased gradually:

- Temperature rises predictably
- Pressure increases slowly and remains tightly constrained
- Safety logic continuously monitors trip thresholds

If the offset is increased too aggressively:

- The PLC will trigger a safety trip
- CalibrationOffset changes are immediately ignored

Operators are expected to **ramp offsets slowly** and observe system feedback.

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9. Why Safety Cannot Be Bypassed

The Helix control system is designed so that:

- Safety variables cannot be directly overridden by operators
- ResetTrip does not force-clear a trip
- Safety logic is evaluated server-side by the PLC

This prevents:

- Accidental unsafe operation
- Malicious misuse of control variables
- Human error during maintenance

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10. Summary

- Trips protect the reactor and are always enforced
- ResetTrip only works when the reactor is genuinely safe
- Maintenance mode allows controlled diagnostics, not full bypass
- The safety window enables limited access without disabling protections

Operators are expected to follow these procedures strictly. Any deviation is logged and audited by the control system.

Helix Industries – Safety First. Always.